Image-assisted Label Connective Completion for Vessel Segmentation with Insufficient Annotations

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***Abstract*—Automatic and accurate vessel segmentation is crucial for disease diagnosis. Deep learning methods are widely used, but their promising results rely on accurately annotated data. Due to complex vessel morphology and low-contrast image, accurate vessel delineation poses a practical challenge, resulting in insufficient annotations, which is a prominent form of noisy labels. This paper proposes an Image-assisted Label Connective Completion method, which enhances label's vessel information by images under the supervision of connectivity to address insufficient annotation issue. Specifically, we develop an Image-guided Vessel Enhancement module, which transmits structural information extracted from images based on label navigation to label space, promoting completion of missing annotated parts in original labels. In addition, a branch completion-connectivity loss is designed and introduced as an auxiliary supervision to prevent vessel branch disconnection during label completion. Experimental results on DRIVE, CHASE_DB1 and DCA1 datasets demonstrate that our method outperforms existing noisy labels learning methods.**

***Index Terms*—Insufficient annotation, Vessel segmentation, Label completion.**

I. Introduction

Vascular diseases are the leading cause of death globally [1], and automatic vessel segmentation that identifies vessel morphological changes is crucial for early screening of many vascular diseases [2], [3]. Facilitated by deep learning methods such as U-Net [4] and its variants [5]–[7], vessel segmentation methods have shown strength in identifying vessel structure from medical images. However, their promising results necessitate a substantial quantity of available pixel-wise accurate annotated data, which are time-consuming to obtain [8]. Moreover, complex vessel morphologies and low-contrast images pose challenges for human annotators, leading to the common situation of insufficiently annotated labels. Therefore, developing vessel segmentation techniques with insufficient annotations is an urgent need.

As insufficient annotations are a prominent form of noisy labels [9], [10], some noisy label learning methods can be possible solutions to handle the insufficient annotation issue. Existing segmentation methods to address noisy labels can be classified into two categories: 1) noisy label suppression and 2) noisy label correction. The first category suppresses noisy labels and drives the segmentation model to focus on clean labels [11]–[14], such as weight control. Though noisy label suppression can reduce the detriment of noisy labels, these approaches overlook rich vessel information like vessel location, distribution and curvature contained in the noisy labels, compromising model generalization. The other category adopts correction strategies to modify noisy labels to reduce the negative impact of wrongly labeled data [15]–[19]. While these approaches demonstrate efficacy in correcting noisy labels, they focus on large targets such as the main vessel stem and neglect to capture variations in vessel details, leading to missing details and disconnected vessels in the results.

Given the insufficient annotations characterizing the noisy labels of vessels, this study proposes completing labels of insufficiently annotated images based on vessel structural information in images. Specifically, we develop an Image-assisted Label Connective Completion ($\mathbf{ILC^2}$) network, which consists of two *(image and label)* branches. Distinct from previous methods that rely solely on images to mine vessel information, $\mathbf{ILC^2}$ utilizes the vessel location information from labels as a navigational guide within images, enabling a more precise extraction of vessel structural information from images. This vessel structural information is then transmitted to the labels, serving to facilitate the completion of the insufficient annotations. Therefore, $\mathbf{ILC^2}$ effectively addresses the challenge of capturing intricate vessel details in low-contrast images and fully explores the rich vessel information contained in labels and images. In the $\mathbf{ILC^2}$ network, an Image-guided Vessel Enhancement (IVE) module is designed to facilitate information transfer and fusion between image and label branches, mitigating potential information aliasing caused by unguided fusion of these data streams. This module further aids the network in precisely identifying and enhancing the vessel features that are present in both the image and the corresponding label. In addition, to address the challenge of over-smoothing of fine structures (scatter of tiny vessels) during label completion, we design an auxiliary supervision called Completion-Connectivity Supervision (CCS) to encourage the completeness and connectivity of each vessel branch.

The contributions of this paper can be summarized as follows: (1) We propose completing vessel labels with enhanced information from images under the supervision of label connectivity to address the insufficient annotation issues in vessel segmentation. (2) We develop an $\mathbf{ILC^2}$ network, which utilizes location-indicated structural information in images to complete missing vessels. A new constraint (CCS) is designed based on

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the connectivity comparison between labels and predictions to enhance the label quality of vessel branches in the completed labels. (3) Experimental results on DRIVE, CHASE_DB1 and DCA1 datasets demonstrate the effectiveness of $\mathbf{ILC^2}$ in completing labels. Our method outperforms the existing noisy label learning methods, achieving stable performance under different annotation insufficiency levels.

II. Method

Following the setting of existing noisy label studies [20]–[22], both samples with clean labels $S_c = \left\{\left(\mathbf{X}_{c(i)}, \mathbf{Y}_{c(i)}\right)\right\}_{i=1}^{N}$ and samples with insufficient annotations *(a form of noisy labels)* $S_n = \left\{\left(\mathbf{X}_{n(i)}, \mathbf{Y}_{n(i)}\right)\right\}_{i=1}^{M}$ are included in the training dataset, denoted as $\mathcal{S} = \{S_c, S_n\}$. $\mathbf{X}_{c(i)}$ and $\mathbf{X}_{n(i)}$ represent the images corresponding to clean and insufficient annotations, respectively. $\mathbf{Y}_{c(i)}$ and $\mathbf{Y}_{n(i)}$ denote the clean and insufficient annotations, respectively. N and M are the sample numbers in S_c and S_n, respectively. We denote the label-completed samples as $S'_n = \left\{\left(\mathbf{X}_{n(i)}, \mathbf{Y}'_{n(i)}\right)\right\}_{i=1}^{M}$. As illustrated in Fig. 1, our method involves three stages: 1) Training the $\mathbf{ILC^2}$ network using S_c; 2) Completing the insufficient annotations by the trained $\mathbf{ILC^2}$ network; and 3) Training segmentation network using S_c and S'_n.

The main contribution of this paper lies in the $\mathbf{ILC^2}$, whose framework is illustrated in Fig. 2. The $\mathbf{ILC^2}$ network includes two encoding streams, one for the original labels and the other for the image-enhanced information. To better fit the morphological features of vessels and facilitate their continuity, the label stream encodes $\mathbf{Y}_c$ with multiple snake convolutions [23], obtaining multi-layer features $\mathbf{F}_l^{(t)}$, where $t \in [1, T]$, T represents the total number of layers in the encoder. We introduce a stack of IVE modules (Sec. II-A), each applied to one layer of label features to provide vessel location information for facilitating image feature extraction, obtaining vessel-enhanced image feature $\mathbf{F}_f'^{(t)}$. The IVE module fuses vessel structural information of images to enhance vessel details in labels, obtaining vessel-enhanced label feature $\mathbf{F}_l'^{(t)}$. An Expanding Set accumulates multi-scale vessel-enhanced label features from previous layers, denoted as $\{\mathbf{F}_l'^{(1)}, \cdots, \mathbf{F}_l'^{(t-1)}\}$. Together with $\mathbf{F}_f'^{(t)}$ initially encoded by the corresponding images, $\mathbf{F}_f^{(t+1)}$ is used as the input of the next layer IVE module, providing enriched information for label completion. Finally, the last layer's output features $\mathbf{F}_f'^{(T)}$ are decoded to get the output labels $\mathbf{Y}'_c$. The whole $\mathbf{ILC^2}$ network is trained with the constraints of a hybrid segmentation loss [24] and a CCS loss (Sec. II-B). Owning to the image assistance and the connectivity constraint, the trained $\mathbf{ILC^2}$ network can assist in completing missing vessels in insufficient annotations while enhancing vessel connectivity in labels.

A. Image-guided Vessel Enhancement (IVE) module

The IVE module exploits the vessel location navigation of labels to facilitate image information extraction, and the vessel structural information of the image is transmitted to enhance vessel information in the label. Taking the IVE module for the t^{th} layer as an example, let $\mathbf{F}_f^{(t)}$ and $\mathbf{F}_l^{(t)}$ be the input image and label features, and denote the corresponding output features as $\mathbf{F}_f'^{(t)}$ and $\mathbf{F}_l'^{(t)}$, respectively. Features in the Expanding Set are element-wise summed, whose average results will be added to $\mathbf{F}_f'^{(t-1)}$ as $\mathbf{F}_f^{(t)}$. $\mathbf{F}_f^{(t)}$ incorporates multi-scale vessel-enhanced label information to accumulatively introduce vessel structural information from images to label features.

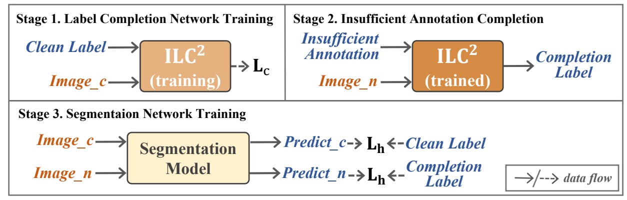

Fig. 1. Three main stages involved in the proposed method.

To evaluate the vessel saliency position for enhancing vessel pixels in label and image, we calculate a vessel-attention matrix $A^{(t)} \in [0,1]$ between feature map $\Phi(\mathbf{F}_l^{(t)})$ and $\Phi(\mathbf{F}_f^{(t)})$, where Φ denotes a feature extraction process. We hope to assign a large weight to features that correspond to vessel to enhance the identification of the detailed parts of vessels. Then, the vessel-attention matrix can be written as

$$A^{(t)} = \theta(\Phi(\mathbf{F}_l^{(t)}) \oplus \Phi(\mathbf{F}_f^{(t)})) \tag{1}$$

where θ and $\oplus$ denote feature extraction and element-wise addition, respectively.

After assigning the weight of $A^{(t)}$ to $\mathbf{F}_l^{(t)}$ and $\mathbf{F}_f^{(t)}$ respectively, we obtain $\mathbf{F}_l'^{(t)}$ and $\mathbf{F}_f'^{(t)}$, which contain knowledgable information about the vessels.

$$\mathbf{F}_l'^{(t)} = \mathbf{F}_l^{(t)} \otimes A^{(t)}, \quad \mathbf{F}_f'^{(t)} = \mathbf{F}_f^{(t)} \otimes A^{(t)} \tag{2}$$

where $\otimes$ represents the element-wise multiplication.

B. Completion-Connectivity Supervision (CCS)

In this paper, we employ a hybrid segmentation loss [24] that exploits gradient smoothing and class balancing [25], [26] as the base supervision for the $\mathbf{ILC^2}$ network, which is defined as

$$L_h = -\frac{1}{P}\sum_{p=1}^{P}(y_p \log y'_p + \frac{2 y_p y'_p}{y_p^2 + y_p'^2}) \tag{3}$$

where $y_p \in Y_c$ and $y'_p \in Y'_c$ denote the label and the corresponding outputs probabilities for the p^{th} pixel.

However, hybrid segmentation loss tends to overlook fine vessels while focusing on the overall overlap, leading to scattered outcomes in completed results. To improve the connectivity of the tree-like vessels during completion, we further design a CCS loss L_{ccs} as auxiliary supervision. The calculation process of L_{ccs} is demonstrated in Fig. 3. We first detect consecutive vessel components by grouping adjacent vessel pixels in $\mathbf{Y}_{c(i)}$ and its corresponding $\mathbf{Y}'_{c(i)}$. Each consecutive vessel component is regarded as a connected branch. The branches extracted from $\mathbf{Y}_{c(i)}$ and $\mathbf{Y}'_{c(i)}$ are denoted as $GB_{i1}, \cdots, GB_{iK}$ and $CB_{i1}, \cdots, CB_{iK}$, respectively, where

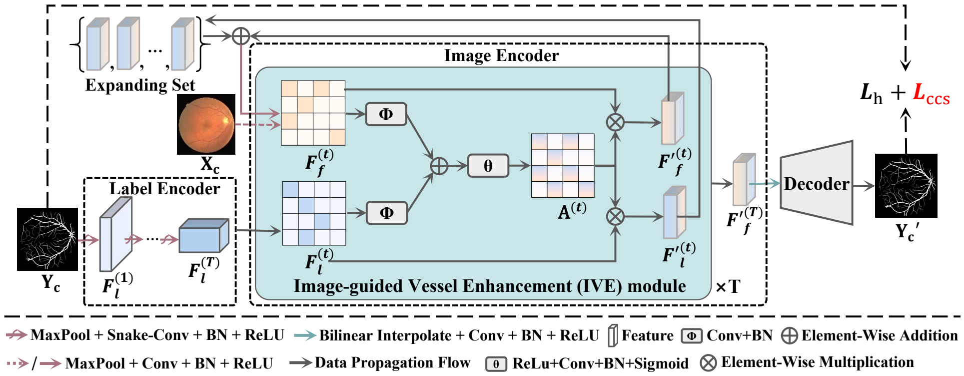

Fig. 2. Illustration of the $\mathbf{ILC^2}$ network. Label features are enhanced by a stack of IVE modules, which are decoded to obtain completed vessel labels.

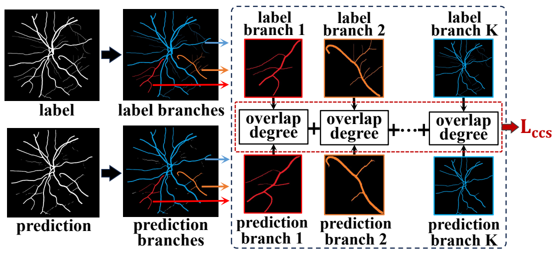

Fig. 3. Illustration of the calculation process of L_{ccs}.

K is the total number of connected branches. The branches with the same subscript k have correspondence, *i.e.*, GB_{ik} *corresponds to* CB_{ik}. We compute the overlap degree of GB_{ik} by CB_{ik} to measure the completeness of each connected branch in the prediction $\mathbf{Y}'_{c(i)}$, written as

$$O_{ik} = \frac{2 \mid GB_{ik} \bigcap CB_{ik} \mid}{\mid GB_{ik} \mid + \mid CB_{ik} \mid} \tag{4}$$

where i and k represent the label and branch index. Summing the overlap degree of all connected branches for samples in S_c, we get L_{ccs}, represented as

$$L_{ccs} = \frac{1}{NK} \sum_{i,k} (1 - O_{ik}) \tag{5}$$

Combining the hybrid segmentation loss and the completion-connectivity loss as completion loss L_c, the total loss for $\mathbf{ILC^2}$ network is

$$L_c = L_h + \lambda L_{ccs} \tag{6}$$

where λ is a balance factor to control L_h and L_{ccs}.

III. Experiments

A. Datasets and Implementation

We validate our method on three reputable public datasets that are widely used in the vessel segmentation community: 1) DRIVE [27] contains 40 retinal images, equally divided by 20:20 for training and testing as in recent studies [3], [28]. 2) CHASE_DB1 [29] contains 28 retinal images, in which the first 20 images are for training, and the remaining 8 images are for testing [30]. 3) DCA1 [31] contains 134 X-ray coronary angiograms, in which the training set consists of 100 images, and the remaining 34 images are for testing [6], [32]. To increase the data diversity, we perform data augmentation by horizontal flip, vertical flip, and rotation in the training phase. To simulate the insufficient annotation situation in practice, we use a Vessel Label Erasing strategy to synthesize an incomplete map by erasing some labeled vessel segments from the ground truth in the training set, following the classical noisy labels setting in [18]. The probability of annotation insufficiency level for each label is selected under a Gaussian distribution, with a mean of μ and a variance of σ^2, to simulate the randomness of insufficient annotations. We set $\sigma^2 = 0.3$ to allow variation of annotation insufficiency and control the values of μ from 0.3 to 0.6, simulating different annotation insufficiency levels in the training dataset. Correspondingly, in these four experiments with different annotation insufficiency level settings, approximately 75%, 84%, 91%, and 95% of training data are insufficiently annotated, and the remaining are clean labels for the $\mathbf{ILC^2}$ network training.

Our experiment involves three phases: 1) $\mathbf{ILC^2}$ network training and insufficient annotation completion, 2) Segmentation model training and 3) Vessel label inference. For the $\mathbf{ILC^2}$ network training, we set the learning rate as 0.01, the batch size as 2, and the total number of encoder layers T as 4 following our baseline. Results in Table I are obtained when λ is 8, which works best for L_c. The trained $\mathbf{ILC^2}$ model is used to complete insufficient annotations. U-Net then is used as the backbone of a vessel segmentation model, with L_h as supervision during training. In the inference phase, the trained segmentation model is applied to predict vessel masks for test data. Evaluation metrics, Dice [25] and IoU [33], are selected to evaluate the vessel segmentation performance, as in [20], [34]. In addition, clDice [35] is introduced as an additional metric in the ablation study to further evaluate the topological connectivity of vessels in completed labels.

B. Results

a) Performance Comparison: Results of our method are compared with four noisy labels learning methods. Among these methods, CoT [13] and MPNN [14] adopt noisy label suppression, while SGL [18] and RMD [19] are noisy label

TABLE I
RESULT COMPARISON ON THE TEST DATA OF DRIVE, CHASE_DB1 AND DCA1 DATASET

Evaluation Criterion	Method	DRIVE				CHASE_DB1				DCA1			
		Annotation Insufficiency Degree				Annotation Insufficiency Degree				Annotation Insufficiency Degree			
		$\mu = 0.3$	$\mu = 0.4$	$\mu = 0.5$	$\mu = 0.6$	$\mu = 0.3$	$\mu = 0.4$	$\mu = 0.5$	$\mu = 0.6$	$\mu = 0.3$	$\mu = 0.4$	$\mu = 0.5$	$\mu = 0.6$
Dice	Baseline (2015)	80.65	79.43	79.02	75.33	79.80	79.11	76.74	75.72	77.93	75.86	73.70	71.96
	CoT (2018)	80.97	78.29	74.31	74.5	79.05	78.69	77.19	74.11	77.66	75.57	73.04	71.37
	SGL (2021)	80.52	77.89	75.74	76.99	80.02	78.44	76.16	75.84	77.91	73.11	72.87	71.10
	RMD (2023)	80.64	78.04	77.63	74.31	79.36	78.36	76.63	75.65	77.21	73.00	71.94	68.95
	MPNN (2024)	78.78	71.68	73.93	65.54	77.36	74.25	70.43	69.36	77.01	71.91	67.99	72.75
	Ours	**81.09**	**81.42**	**81.06**	**79.54**	**80.18**	**80.27**	**79.16**	**77.46**	**78.20**	**76.86**	**74.73**	**74.65**
IoU	Baseline (2015)	67.61	65.94	65.36	60.48	66.45	65.49	62.32	61.04	64.11	61.56	58.66	56.81
	CoT (2018)	68.05	64.41	59.29	59.67	65.41	64.9	62.88	59.01	63.71	60.94	57.85	56.05
	SGL (2021)	67.49	63.84	61.22	62.66	66.73	64.55	61.53	61.16	63.94	57.78	57.47	55.30
	RMD (2023)	67.61	64.06	63.50	59.27	65.82	64.46	62.17	60.87	63.07	57.70	56.68	53.21
	MPNN (2024)	65.08	56.11	58.94	49.05	63.16	59.10	54.43	53.18	62.82	56.78	52.63	57.59
	Ours	**68.24**	**68.70**	**68.23**	**66.28**	**66.95**	**67.08**	**65.54**	**63.28**	**64.38**	**62.57**	**59.97**	**59.84**

TABLE II
LABEL COMPLETION ABILITY EVALUATION IN THE ABLATION EXPERIMENT ON DRIVE DATASET

Evaluation Criterion	Method	Annotation Insufficiency Degree			
		$\mu = 0.3$	$\mu = 0.4$	$\mu = 0.5$	$\mu = 0.6$
Dice	Baseline	81.09	80.50	80.00	76.73
	Baseline+IVE	81.70	80.78	80.33	77.04
	Baseline+IVE+L_{ccs}	**82.87**	**81.48**	**80.78**	**77.71**
IoU	Baseline	68.26	67.44	66.91	62.89
	Baseline+IVE	69.14	67.86	67.36	63.22
	Baseline+IVE+L_{ccs}	**70.79**	**68.83**	**68.01**	**64.01**
clDice	Baseline	80.87	80.06	79.40	76.45
	Baseline+IVE	81.69	80.16	80.20	76.74
	Baseline+IVE+L_{ccs}	**82.63**	**81.70**	**80.48**	**77.80**

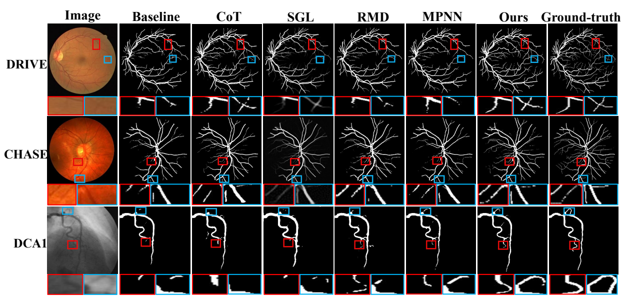

Fig. 4. Examples of segmentation results on test data.

correction methods. U-Net is shown as baselines, and we re-implement all compared methods with the same U-Net backbone for fair comparison. Comparison results on test data are presented in Table I. It is shown that our method achieves the best results on all evaluation criteria for cases with different annotation insufficiency levels. The performance improvement is more conspicuous when the labels have a higher annotation insufficiency level. In addition, we find that the results of our method are relatively stable in different annotation insufficiency levels, superior to the fluctuated performance of other methods. We further illustrate vessel segmentation results in Fig. 4, validating the qualitative superiority of our method.

b) Ablation Study: To validate the effectiveness of the IVE module and CCS loss, this section shows the results of ablation studies that disable each component when learning the $\mathbf{ILC^2}$ network in the DRIVE dataset. We assess completed label $\mathbf{Y}'_n$ compared to corresponding ground truth *(unpoisoned label)* to evaluate the completion ability of the $\mathbf{ILC^2}$ network.

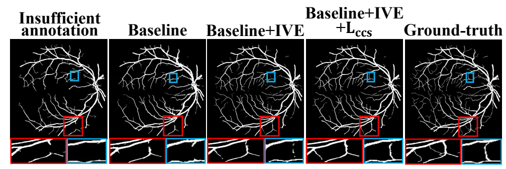

Fig. 5. Label completion ability visualization of ablation study on DRIVE.

The unpoisoned labels of S_n are only used as references for label completion evaluation. The network trained by the hybrid segmentation loss L_h is considered as baseline. As shown in Table II and Fig. 5, label completion results are improved when adding the IVE module to the baseline, indicating that the IVE module effectively blends vessel information from the image into the label. Enhancement in completed vessel connectivity can also be found with the addition of L_{ccs}, reflecting the necessity of topological connectivity preservation during vessel label completion. In total, the best results are achieved with both the IVE module and CCS loss.

IV. CONCLUSION

This paper develops an $\mathbf{ILC^2}$ network that completes insufficient annotations with the assistance of images to address the insufficient annotation issues in vessel segmentation. We design an IVE module to facilitate information transition from image to label. In addition, a CCS loss is designed as an auxiliary supervision to guide $\mathbf{ILC^2}$ network to preserve connectivity and integrity during label completion. Experimental results on three vessel datasets demonstrate that $\mathbf{ILC^2}$ outperforms existing noisy label learning methods. We superimpose the IVE module and CCS loss and see a positive effect of them in label completion. In the future, we will strive to further extend application scenarios in the tubular structure of our method.

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